Ivan Danielov Ivanov

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Summary

Data scientist and ML engineer with over 10 years of professional experience building impactful products, leading technical teams, delivering scalable ML solutions, and navigating early-stage startup dynamics.

Skills

Technical Leadership | Product Discovery | Machine Learning | AI Engineering | LLM Orchestration | RAG | AI Security & Red Teaming | Python | SQL | Vector Databases | Time-Series Analysis | Anomaly Detection | Predictive Maintenance | Recommendation Systems | Knowledge Graphs | Big Data | Google Cloud Platform | Microsoft Azure | REST APIs | CI/CD | Sales Engineering | Public Speaking

Professional Experience

Stihia.ai - Full-Stack and AI Engineer, Founder

Sofia, Bulgaria | September 2025 - Present

- Founded an AI security company as a solo founder, focused on red teaming and real-time threat detection for AI agents, and later brought in 3 senior experts in cybersecurity, product management, and business development as co-founders.
- Conducted over 50 customer discovery interviews for product validation with senior leaders from banks, fintech scaleups, and multiple startups.
- Developed a scalable multi-tenant application for OpenTelemetry-based AI security observability and real-time prompt-injection detection in agentic systems (Python, Next.js, GCP, Vertex AI, open-source LLMs). Built a scalable FastAPI-based public backend API service. Implemented guardrail techniques, including both deterministic (e.g., regex) and probabilistic (e.g., LLM-as-a-judge) approaches, primarily for banking and fintech use cases.
- Published Stihia Zmey, a capture-the-flag cybersecurity game for LLM prompt-injection practice that has been played 2,000 times by anonymous users.
- Built multiple demos for customers to showcase AI agent vulnerabilities using ReAct agents, tools, and diverse LLMs (using LiteLLM as a unified API gateway).
- Independently explored and experimented with adversarial examples and security methodologies for agentic AI systems.
- Delivered multiple lectures, webinars, and workshops on AI security, including for the Bulgarian Academy of Sciences, Sofia University "St. Kliment Ohridski", SoftUni AI, and various podcasts.

Ethermind - Chief Technology Officer

Sofia, Bulgaria | November 2023 - August 2025

- Defined the product strategy, architecture, and roadmap; hired and mentored a high-performing team of 5 engineers to build the startup's MVP, enabling GBP 500K in pre-seed funding from a Bulgarian VC fund and multiple angel investors.
- Conducted customer discovery interviews and participated in all sales meetings as a technical domain expert and sales engineer.
- Led the development of a scalable agentic LLM-based RAG system (Python, LangChain, Microsoft Azure, PostgreSQL, pgvector, Chroma, OpenAI, Azure OpenAI Service, Vertex AI, Gemini, ElevenLabs) and a multi-tenant application (Node.js, React, PostgreSQL, Auth0) serving over 300 business users and helping them interpret unstructured enterprise data.
- Structured the development process around GitHub Actions, Docker containerization, and Terraform infrastructure-as-code for Microsoft Azure.
- Continuously improved retrieval quality, semantic search relevance, and answer grounding for enterprise-specific knowledge bases based on interaction observability and golden-dataset evaluations.
- Continuously optimized the team's software development processes by combining agile methodologies (experimenting between Scrum and Kanban), continuous delivery, and AI-assisted development tools (GitHub Copilot, Cursor, Windsurf).
- Managed the IT and cybersecurity operations of the company, implementing controls for ISO 27001 compliance.

Humans in the Loop - Chief Technology Officer

Sofia, Bulgaria | November 2022 - June 2023

- Defined the technology strategy, architecture, and roadmap for the social-impact business, helping underprivileged people from conflict-affected countries access fairly paid data annotation work for computer vision projects.
- Conducted B2B customer discovery interviews and developed the MVP of a platform for live human oversight of AI systems, ensuring work for 4 refugees based in Sofia, Bulgaria, annotating a real-time image feed for exception handling in autonomous forklifts.
- Built and productionized the platform using Python (Flask), PostgreSQL, GCP, Docker, Terraform, and Label Studio.
- Managed the IT and cybersecurity operations of the company, hired an IT administrator, and implemented controls for SOC 2 and ISO 27001 compliance.

SoftServe - Lead Data Scientist

Sofia, Bulgaria | January 2019 - November 2022

- Joined the Center of Excellence's Data Science Group as an internal Data Science consultant for various company-wide machine learning projects and initiatives.
- Worked as a hands-on Data Scientist, applying data analysis (EDA, feature engineering, reporting), data engineering (Apache Beam, Apache Airflow, BigQuery), machine learning (Random Forest, Gradient Boosting, Transformers, NLP, time-series forecasting, recommendation systems), and software engineering (Python, Docker, CI/CD, Google Cloud Platform, Microsoft Azure).
- Acted as sales engineer and Data Science domain expert (equivalent to a forward-deployed engineer) for multiple pre-sales activities, including on-site discovery workshops with US-based enterprises in Texas, Michigan, and Illinois. Translated vague customer requirements into specific ML solution architectures with measurable outcomes.
- Acted as tech lead for Data Science teams of up to 6 people in both short- and long-term projects.
- Acted as technical interviewer in the recruitment process for the Data Science Center of Excellence.
- Notable projects:
  - Teaching personalization for US school districts covering over 40,000 students. The solution used knowledge graphs, transformer models, collaborative-filtering recommendation systems, and Google Cloud Platform (BigQuery, Dataflow, Composer). Led a team of 5 data scientists within a cross-functional team of 30 people and worked closely with the product manager on roadmap management. Designed a structured and iterative delivery process that ensured an outcome-oriented research phase followed by a productionization phase for all ML features. The process proved to be reliable with over 7 ML features being delivered to production and continuously improved for 9 months.
  - Predictive maintenance of cryopumps used in leading semiconductor manufacturing fabs worldwide, based on time-series classification using random forest models and scalable MLOps pipelines in Microsoft Azure (Azure ML, SQL, Anomaly Detector). Developed automated MLOps processes for continuous model training, evaluation, versioning, and deployment, with model-quality and data-drift tracking over time.
  - Demand prediction and fraud detection for large-scale US retail chains with millions of daily transactions using LightGBM models and Google Cloud Platform (Dataproc, BigQuery, Composer).
  - Automated extraction of structured data from German medical documents using intelligent document processing and fine-tuned spaCy named entity recognition (NER) models.

Promotions:

- **Lead Data Scientist:** February 2021 - November 2022
- **Data Scientist:** January 2019 - January 2021

Progress Software (ex. Telerik) - Data Scientist

Sofia, Bulgaria | November 2016 - December 2018

- Joined the Business Intelligence team as the first Data Scientist in the company, providing senior management with insights into past, current, and future business performance based on data collected by various business systems (Salesforce, Oracle Eloqua, Google Analytics, etc.).
- Developed internal Machine Learning applications that helped the sales and marketing teams improve performance while managing thousands of leads per day.
- Built scalable Python infrastructure for daily live predictions integrated directly into Salesforce, including data cleaning, feature engineering, cross-validation, grid search, and explainability factors for robust model performance.
- Notable projects:
  - Scoring potential buyers for their propensity to purchase (Lead Scoring) using a logistic regression model with an explainability module for trust in model decisions.

- Scoring existing clients for their propensity to churn (Attrition Scoring) using decision tree models.
- Customer segmentation for marketing purposes using K-Means and Agglomerative Clustering.

Promotions:

- **Data Scientist:** March 2018 - December 2018
- **Data Scientist, Associate:** February 2017 - February 2018
- **BI Performance Analytics Intern:** November 2016 - February 2017

Recogizer GmbH - Data Science Intern

Bonn, Germany | August 2016 - October 2016

- Implemented scalable Python data pipelines for collecting and processing real-time IoT sensor telemetry from smart, green buildings in an energy-efficiency context using Apache Kafka and Apache Cassandra.

University of Bonn, Knowledge Discovery and Machine Learning Research Group - Student Assistant

Bonn, Germany | October 2015 - September 2016

- Tutored exercise groups for the graduate lectures "Learning from Non-Standard Data" and "Data Mining and Knowledge Discovery" while conducting research for my master's thesis.

University of Bonn, Autonomous Intelligent Systems Group - Student Research Assistant

Bonn, Germany | October 2014 - June 2015

- Developed C++ software modules for the NimbRo Rescue robot using ROS, including a network-condition simulator and a 3D camera visualization module for VR-based teleoperation.
- Contributed to the team's **4th-place** finish (out of 23 worldwide teams) at the DARPA Robotics Challenge 2015 in Pomona, California.

MicroAccount - Software Developer

Varna, Bulgaria | July 2012 - August 2013

- Developed web and cross-platform mobile applications for the agricultural industry using JavaScript, C#, and PhoneGap.
- Built custom SugarCRM modules in PHP, including a guard-shift scheduler that generated constrained monthly schedules using a genetic algorithm.

Education

University of Bonn - Master of Science in Computer Science

Bonn, Germany | October 2013 - October 2016

- **GPA:** 1.9 (1 is best, 5 is worst)
- **Main subjects:** Machine Learning, Data Mining, Neural Networks, Data Science, Big Data, Algorithms, Distributed Systems, Robotics, Semantic Web.
- **Master's thesis:**
  - **Topic:** Feature Extraction from Bounded Tree-Width Graphs using Canonical String Representations.
  - **Overview:** Developed a novel method for feature extraction from graphs that can be used as a graph kernel for any kernel method (e.g., SVM) to solve graph learning problems.
  - **Empirical evaluation:** Tested the algorithm empirically on classification problems using well-known chemical compound datasets (Mutagenesis, Carcinogenesis, NCI-HIV) and category recognition in the DBpedia RDF dataset.

Technical University of Varna - Bachelor of Engineering in Computer Engineering

Varna, Bulgaria | September 2009 - July 2013

- **GPA:** 5.85 (6 is best, 2 is worst)
- **Main subjects:** C++ and Java Programming, Web development, Microprocessor Systems, Calculus, Electronics
- **Bachelor's thesis:** Melody prediction - given a single-track digital melody in MIDI format, the application used a neural network (a multi-layer perceptron written in Java) to generate a continuation of the melody.

Side Activities

Labsi - Data Scientist, Co-Founder

Sofia, Bulgaria | May 2022 - Present

- Built applied AI capabilities for a health-tech mobile application that provides scalable storage, processing, and analysis of laboratory test results, with 1,000+ users and 3,000+ processed documents.
- Led document-intelligence work using Python, Google Cloud Platform (Vertex AI Document AI, Gemini, BigQuery) to extract structured indicators from noisy medical documents and enable trend tracking over time.
- Shaped early system architecture, analytics setup, usage metrics, and product strategy under typical early-stage startup constraints.
- Productionized the application on Google Cloud Platform using CI/CD, Docker, and Terraform, and helped publish the mobile app on the Apple App Store and Google Play Store.
- Contributed to compliance and risk mitigation around GDPR, security, and health-data sensitivity.

Data For Good - Bulgaria (NGO) - Data Science Volunteer, Board Member

Sofia, Bulgaria | February 2021 - Present

- Co-founded the organization to address environmental, educational, and democracy-related challenges for the public benefit using data work.
- Helped build a community of 89 volunteers for the organization.
- Contributed to various projects, including a dashboard for government spending data visualization and a report about high-risk sections for fraud in Bulgarian parliamentary elections.

Sofia University "St. Kliment Ohridski", Faculty of Mathematics and Informatics - Guest Lecturer

Sofia, Bulgaria | February 2026 - June 2026

- Co-deliver the course *Large Language Models and their Applications*, lecturing on hallucinations, RAG, AI agents and tool use.

Project Stork (Bulgarian Open-Source LLM) - Data Engineering Volunteer

Remote | June 2023 - January 2025

- Designed and implemented the data-processing pipeline for hundreds of gigabytes of Bulgarian text from books, news, social media, and web sources.
- Built scalable cleaning, normalization, and tokenization workflows in Apache Beam running on Google Cloud Dataflow.

Publications

- *DRC Team NimbRo Rescue: Perception and Control for Centaur-like Mobile Manipulation Robot Momaro* | DARPA Robotics Challenge Finals book (Wiley, originally Journal of Field Robotics) | November 2016

Certifications

- Google Cloud Certified Professional Data Engineer | Google Cloud | September 2020 (expired September 2022)

Languages

- English | Fluent
- German | Basic (B1 Certificate)
- Bulgarian | Native